



AUTOMATED EXTRACTION AND TOPOLOGICAL VALIDATION OF URBAN ROAD NETWORKS FROM OPENSTREETMAP

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Automated Extraction and Topological Validation of Urban Road Networks from OpenStreetMap

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Reliable traffic data is crucial for effective transport planning and environmental monitoring, yet official datasets such as the UK Department for Transport’s Annual Average Daily Traffic (AADT) counts tend to emphasise major roads while overlooking minor and residential streets. This study addresses the gap by integrating AADT with OpenStreetMap (OSM), creating a harmonised and machine learning-ready dataset for Newcastle upon Tyne. Using geospatial processing, AADT count points were matched to the OSM road network through a quality-controlled nearest-neighbour approach, ensuring accurate alignment between observed traffic flows and network geometry. Feature engineering enriched the dataset with attributes such as road length, classification encodings, and spatial proximity, enabling prediction of traffic volumes on unmonitored roads. The findings highlight the importance of minor roads, which dominate urban networks but are often excluded from official monitoring. The proposed workflow demonstrates a scalable and transferable method that can support more comprehensive traffic modelling, accessibility studies, and pollution analysis in other cities.

1 Introduction

Urban mobility has emerged as one of the defining challenges of 21st-century cities. Traffic congestion, deteriorating air quality, and unequal accessibility impact public health, economic productivity, and environmental sustainability [3, 5]. Addressing these challenges requires robust, fine-grained data on traffic flows across the full hierarchy of road networks. In the United Kingdom, the Department for Transport (DfT) provides Annual Average Daily Traffic (AADT) counts at selected monitoring points, which remain the gold standard for traffic measurement. However, AADT coverage is biased toward motorways and major A-roads, leading to underrepresentation of local and residential streets that nevertheless account for a large share of daily trips and emissions [13]. This imbalance reduces the ability of planners and researchers to design interventions that address neighbourhood-level congestion, pollution hotspots, or equity of access.

Parallel to these challenges, the emergence of Volunteered Geographic Information (VGI) has transformed the availability of spatial datasets for transport research. OSM [28] in particular has developed into the most comprehensive open-access road dataset globally, with applications spanning navigation, routing, disaster response, and environmental modelling [1, 4]. OSM provides rich geometric detail and hierarchical road classifications, but as a crowdsourced platform, its quality varies geographically and thematically. Positional accuracy in urban cores can rival official datasets, while completeness in rural or less-mapped areas remains inconsistent [12, 17, 19]. Attributes such as lane counts, speed limits, and surface types are particularly prone to missing values [6].

Advances in machine learning (ML) have significantly improved traffic modeling. Traditional regression-based AADT estimation often fails to capture the non-linear relationships between traffic flows, land use, and network structure. Deep learning architectures—including combinations of linear models and activation layers—have demonstrated the ability to model sudden traffic regime changes more accurately than classical approaches [30]. Tree ensemble methods, especially XGBoost, offer rapid and interpretable solutions that outperform both support vector machines and neural networks, particularly when integrating graph-based spatial features [11]. Broad surveys of AI-driven traffic models further

highlight the flexibility and precision of modern ML techniques, such as regression trees, random forests, and boosting algorithms, over traditional statistical methods [2, 8].

Despite these advances, there remains a disconnect between authoritative AADT datasets and VGI-derived road networks. Most UK studies use DfT counts in isolation or apply OSM primarily for routing and simulation [7, 34]. Few attempts have been made to systematically integrate the two sources into a unified, ML-ready dataset, especially for medium-sized cities such as Newcastle upon Tyne, where minor roads are heavily used but sparsely monitored. This dissertation addresses that gap by aligning OSM geometries with DfT count points, applying rigorous spatial validation, and engineering predictive features to create a harmonised dataset for traffic modelling.

The research is structured around four objectives:

- To conduct exploratory data analysis (EDA) of OSM and DfT AADT datasets, identifying coverage and patterns for major and minor roads.
- To preprocess both datasets, addressing missing geometry, filtering irrelevant features, and harmonising coordinate reference systems
- To integrate AADT count points with OSM road edges through spatial joins, validating nearest-neighbour matches and applying quality thresholds.
- To engineer features such as road length, classification encodings, and distance measures, producing a dataset suitable for ML-based traffic prediction.

The contribution lies in demonstrating a replicable workflow that bridges authoritative traffic counts and VGI road geometries, with a specific focus on minor roads in Newcastle. By addressing the underrepresentation of these roads in traditional datasets, the project supports more comprehensive traffic modelling, pollution analysis, and urban accessibility research.

2 Literature Review

2.1 OpenStreetMap and VGI in GIScience

OSM is the leading example of VGI, where contributors collaboratively digitise roads, buildings, and points of interest [4]. Its openness has enabled widespread use across academic and industrial domains, but quality varies with geography and contributor activity [22]. Studies comparing OSM to authoritative datasets report generally high positional accuracy in urban cores but lower completeness in rural or under-mapped regions [17].

[4] also highlighted OSM’s rapid growth in GIScience applications, while [40] demonstrated how participatory mapping processes influence coverage and reliability. [32] pointed to logical inconsistencies such as network disconnections, while [34] explored the benefits of hybrid datasets combining OSM with official road data. These findings underline OSM’s value as a global, flexible database, but also the necessity for quality checks before integration into modelling workflows.

2.2 Methods for Improving OSM Data Quality

Numerous techniques have been proposed to address OSM’s limitations. [18] introduced a network-based hierarchical feature augmentation approach to refine road classification predictions, while [12] developed QualiOSM to enrich attribute completeness by aligning OSM with external datasets, demonstrated in Prague. Error-detection frameworks such as [41] validate geometry integrity, whereas [33] showed the benefits of storing OSM networks in graph databases for consistency.

Systematic evaluation frameworks and rule-based logical checks [32] further improve reliability. Collectively, these contributions establish a toolkit for preprocessing and validating OSM networks, which this dissertation leverages in preparing ML-ready datasets.

2.3 OSM in Urban and Environmental Applications

OSM has proven adaptable across diverse urban and environmental domains. [16] compared OSM cycling infrastructure with municipal open data, demonstrating high overlap but revealing missing links. [27] applied OSM networks to simulate pedestrian routes to green spaces, while [29] evaluated OSM food-related POIs for epidemiological research. In climate and environmental studies, OSM has supported climate risk assessments [25], noise pollution mapping [31], and renewable energy research such as wind turbine siting [23].

Large-scale applications include [39] global OSM-derived urban network dataset, and [10] generation of transit maps. Even in remote regions such as the Arctic, [24] tested OSM for industrial site mapping, again finding variability across sparsely mapped zones. These applications highlight OSM’s versatility, but also its dependency on region-specific validation.

2.4 OSM in Transport and Traffic Modelling

Transport is one of OSM’s core application areas. [7] demonstrated OSM’s use in SUMO-based simulations of Bologna’s ringway, while [26] created routing systems optimised for pedestrian comfort. OSM has also supported autonomous driving research, with von [35] using it for validating semantic segmentation models. Machine learning approaches further expand OSM’s potential. [36] extracted multilane roads using graph convolutional neural networks, while [9] advanced semi-supervised learning on sparse OSM features. Such works reinforce OSM’s ability to underpin advanced transport models, provided preprocessing ensures logical integrity.

2.5 AADT Estimation and Machine Learning Approaches

AADT remains central to traffic planning, yet count coverage is limited. Traditional regression methods have struggled to extrapolate flows on minor roads. [21] showed that machine learning—specifically random forests and deep learning—outperform regression by capturing nonlinear patterns. Recent studies suggest that adding spatial and contextual variables improves prediction accuracy. [18] demonstrated that hierarchical network features enhance classification, while [37] linked OSM-derived variables to regional economic evaluations. [14] applied OSM to assess territorial innovation, and [20] validated OSM’s suitability for navigation in Australian road networks. These findings suggest OSM and AADT integration holds significant potential for ML-based traffic prediction.

2.6 Identified Gaps

While OSM’s role in traffic modelling is increasingly recognised, systematic integration with AADT remains limited. Most studies either focus on OSM data quality in isolation [22, 42] or apply ML to AADT without leveraging VGI-derived road geometries [21, 38]. Hybrid examples such as [7] and [34] illustrate complementarity but stop short of ML-ready integration. Furthermore, few works examine mid-sized UK cities such as Newcastle, where coverage gaps in minor roads persist. This dissertation addresses this gap by aligning OSM geometries with DfT count points, validating spatial joins through distance thresholds, and engineering predictive features. The resulting integrated dataset provides a foundation for

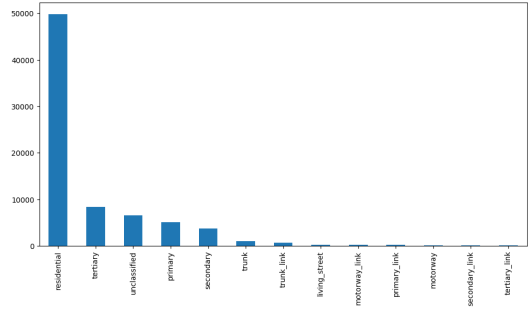


Fig. 1. OSM Road Type Distribution..

future ML models aimed at estimating minor road traffic, an area underrepresented in existing research.

3 Methodology

The methodology was designed to create a reproducible, scalable pipeline for integrating OSM and DfT AADT datasets. The study area was defined as Newcastle upon Tyne, UK, chosen for its urban complexity and availability of DfT monitoring points.

Data acquisition and preprocessing began with the extraction of the drive network from OSM using the OSMnx library. The unsimplified network yielded 76,121 edges and 40,045 nodes, with attributes such as `highway`, `length`, and `geometry` retained for analysis. Summary statistics for the extracted dataset are reported in Table 1. To normalise attribute inconsistencies, `highway` values stored as lists were converted to string format. Road types were classified into major (motorway, trunk, primary, secondary) and minor (residential, tertiary, unclassified, service) classes. Only these target types were retained, while edges with missing geometry were discarded.

Metric	Value
Total nodes	40,045
Total edges	76,121
Distinct highway types	13
Attribute coverage: lanes	8.9%
Attribute coverage: maxspeed	39.0%
Attribute coverage: bridge/tunnel	0.8%
CRS	EPSG:4326

Table 1. Summary of road network metrics

Figure 1 illustrates the distribution of road segments across OSM classes, confirming the dominance of residential roads in Newcastle. These streets not only represent the largest share of segments but also account for the greatest proportion of total network length, underscoring their importance in local traffic dynamics.

The DfT dataset consisted of 2,587 AADT records linked to 211 geolocated count points. These datasets were merged using the `count_point_id` attribute, producing a comprehensive table of traffic volumes disaggregated by vehicle type. Temporal variation in traffic

demand was explored by aggregating volumes across years, while spatial coverage was validated against OSM geometries.

Spatial integration was conducted through a nearest neighbour join. Both datasets were reprojected to the British National Grid (EPSG:27700) to ensure accurate distance computation. For each count point p , the nearest road segment r^* was selected according to:

$$r^* = \arg \min_{r \in R} d(p, r) \quad (1)$$

where $d(p, r)$ is the Euclidean distance between point p and candidate segment r in projected space.

To reduce noise, countpoints more than 50 m from the nearest road were excluded, following best practice thresholds for spatial matching. After filtering, 3,164 valid point–road matches were retained. Results of this quality assurance process are summarised in Table 2.

Step	Count of Points
Initial matched points	3,326
Points > 50 m from nearest road	162
Valid points retained after filtering	3,164

Table 2. Point filtering summary

Feature engineering was performed to produce a machine-learning-ready dataset. Each road segment was assigned its geometric length, calculated as:

$$L_r = \sum_{i=1}^n \|v_{i+1} - v_i\| \quad (2)$$

where v_i are vertices along the polyline geometry of road r . Road types were one-hot encoded (e.g., `type_residential`, `type_primary`), and additional features included the nearest neighbour distance between AADT points and OSM edges.

The machine-learning dataset contained 64,207 rows and 10 features, with the dependent variable defined as `all_motor_vehicles` from the AADT dataset.

4 Results

The OSM extraction for Newcastle produced a dense and highly connected road network, with 76,121 edges and 40,045 nodes. However, attribute completeness was uneven: while highway type and geometry were universal, lane count (13 coverage), maximum speed (45), and bridge/tunnel attributes (<1) were sparsely populated. These results are reported in Table 1, consistent with previous findings that OSM prioritises core topology while secondary attributes remain incomplete [6, 15]w.

Road type distribution (Figure 1) showed that residential streets dominate the network (49,792 segments), corresponding to over 1,289 km of total road length. By contrast, motorways (123 segments, 3.8 km) and trunk roads (1,007 segments, 56.5 km) accounted for only a small proportion of the total. These findings underscore the importance of minor roads in shaping Newcastle’s mobility and environmental impacts.

Integration with the AADT dataset began with 2,587 records and 211 count points. Road type distribution within the countpoints dataset was skewed towards minor roads (121 locations) compared with major (90), highlighting its value for studying underrepresented segments. Temporal analysis revealed fluctuating traffic demand Figure 2, including declines

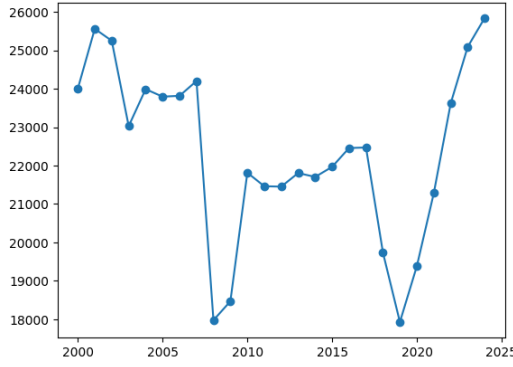


Fig. 2. Temporal Traffic Trends (2005–2024).

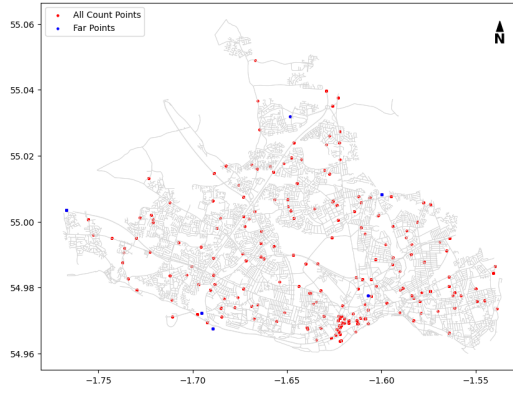


Fig. 3. AADT Points >30 m from Nearest Road.

during the 2008–2010 financial crisis and 2019–2020 pandemic, before recovering to over 26,000 vehicles/day by 2024. Traffic heterogeneity across road hierarchies for minor roads are substantially lower than for major roads.

Spatial integration produced 3,326 initial matches, with an average offset of 9.0 m from OSM segments. Quality assurance identified 162 mismatches exceeding 50 m, visualised in Figure 3, where blue points represent anomalous count locations (typically at multi-level interchanges). After filtering, 3,164 validated matches were retained (Table 2).

The final integrated dataset contained 64,207 road segments with 10 engineered features, including geometric measures, nearest-neighbour distances, and categorical encodings. This dataset establishes a strong foundation for machine learning approaches, enabling traffic prediction on unmonitored links while capturing the heterogeneity of Newcastle’s urban road hierarchy.

5 Conclusion

This study developed and validated a reproducible pipeline for integrating OSM and AADT datasets in Newcastle upon Tyne. The methodology successfully extracted, filtered, and spatially aligned authoritative traffic counts with volunteered geographic information, producing a machine-learning-ready dataset. Results confirmed that residential and tertiary

roads dominate the urban network, both in frequency and extent, yet remain underrepresented in traditional monitoring.

The study also highlighted key limitations. Attribute completeness in OSM was inconsistent, with lanes and maximum speed data missing for a majority of edges. Spatial mismatches between AADT points and OSM geometries reached up to 163 m in rare cases, underscoring the importance of rigorous quality assurance in future work.

Despite these limitations, the integrated dataset offers a robust basis for predictive traffic modelling. Future work should expand the feature space with contextual variables such as urban/rural classification, land use from the Urban Atlas, and population density, which would improve model accuracy by linking road segments with socio-spatial drivers of demand. Advanced modelling approaches such as graph neural networks (GNNs) or spatial autoregressive models could further leverage the topological structure of road networks. Longitudinal analyses incorporating policy interventions—such as low-traffic neighbourhoods (LTNs) or clean air zones—would extend the utility of this dataset to traffic management and environmental planning.

In doing so, this work provides both a methodological framework and a transferable dataset model, contributing to more inclusive and evidence-based transport planning that fully accounts for the role of minor roads in urban mobility systems.

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